

# **Internship Project Phase 2 Report (BUILD)**

**Name: Ashwin R Vasistha**

**USN: 1BO22CS006**

**College: Brindavan College of Engineering**

**Branch: CSE**

**e-mail: ashurashwin@gmail.com**

**Project: E-Commerce Sales Performance Dashboard with Cohort Analysis**

**External guide:**

- 1. Rakshit Shetty – Director, Rooman Technologies**
- 2. Shruthi M – Mentor , Rooman Technologies**

## 1. Introduction

Phase 2 focuses on transforming raw transactional data into meaningful insights using Python. This phase involves data cleaning, exploratory data analysis (EDA), and cohort analysis to understand customer behavior and sales performance.

## 2. Objectives

- To clean and preprocess the dataset for accurate analysis
- To perform exploratory data analysis to identify trends and patterns
- To analyze customer behavior using cohort analysis
- To generate insights for business decision-making
- To prepare data for dashboard visualization in Power BI

## 3. Dataset Description

The dataset used is the **E-Commerce Customer Behaviour**, which contains transactional data from a retail store.

### Key attributes include:

- Order: Unique transaction ID
- Category and Sub-category
- Description: Product name
- Quantity: Number of items purchased
- Order Date
- Total Amount
- State and City: Customer location

## 4. Data Cleaning and Preprocessing

Data cleaning was performed to ensure accuracy and reliability of analysis.

### Steps performed:

- Combined List of Orders and Order Details
- Removed rows with missing Customer IDs
- Removed duplicates
- Converting Data Types

- Creating Customer Identifier
- Handling Negative Profit (Not Removing)
- Create Time Features.

After cleaning, the dataset contained only valid and meaningful transactions.

## **5. Feature Engineering**

New features were created to enable time-based analysis:

- Cohort Month → Monthly grouping of transactions
- Year → Year of transaction
- Month → Month number

These features helped in analyzing trends and customer behavior over time.

## **6. Exploratory Data Analysis (EDA)**

EDA was performed to understand sales performance and customer patterns.

### **6.1 Monthly Revenue Trend**

Revenue showed a consistent increasing trend with a peak towards the end of the year, indicating seasonal demand during holiday periods.

### **6.2 Top Products Analysis**

A small number of products contributed significantly to total revenue, indicating a skewed sales distribution.

### **6.3 Region-wise Sales Analysis**

Some regions generate high revenue but low or negative profit, indicating cost inefficiencies.

Sales are concentrated in a few key states, contributing the majority of revenue.

### **6.4 Orders and Customer Growth**

Both the number of orders and unique customers increased over time, indicating business growth and expanding customer base.

## 6.5 Revenue Distribution

Revenue distribution was highly skewed, with most transactions being low value and a few high-value purchases contributing significantly.

## 7. Cohort Analysis

Cohort analysis was performed to study customer retention over time.

Customers were grouped based on their **first purchase month (Cohort Month)**, and their activity was tracked across subsequent months.

### 7.1 Methodology

- Identified first purchase month for each customer
- Calculated time difference between purchases (Cohort Index)
- Created a retention matrix
- Visualized retention using a heatmap

### 7.2 Key Findings

- Customer retention is highest during the first month
- Retention decreases gradually over time
- Some cohorts show better retention, indicating strong customer engagement during certain periods

## 8. Output Generated

- Cleaned dataset (cleaned\_data.csv)
- Cohort retention dataset (cohort\_retention.csv)
- Visualizations for trends and patterns

These outputs were used for dashboard creation in the next phase.

## 9. Conclusion

Phase 2 successfully transformed raw data into actionable insights. The analysis revealed strong sales growth, seasonal trends, product concentration, and customer retention patterns. These findings provide valuable input for business decision-making and dashboard visualization.